Vishal Vaka

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SUMMARY

Software Engineer with 2+ years building production systems and scalable backend infrastructure. Strong foundation in data structures, algorithms, and system design. Experienced in full software development lifecycle from design through deployment with focus on code quality, testing, and cross-functional collaboration.

SKILLS

- **Programming Languages**: Python, C, C++, Java, SQL, Bash
- **Backend & Distributed Systems**: REST APIs, Spring Boot, Node.js, Microservices, Distributed system design, Redis, Docker, Kubernetes
- **Databases & Storage**: PostgreSQL, MySQL, MongoDB, Redis, Database Design, Query Optimization, Indexing
- **Cloud & Tools**: AWS (Lambda, S3, EC2, SQS, SNS), Git/GitHub Actions, Jenkins, CI/CD, Splunk
- **Core CS**: Data Structures, Algorithms, System Design, OOP, Multithreading, Debugging, Performance Optimization

EDUCATION

University of Illinois at Chicago **Aug 2023 - May 2025**

Master of Science, Computer Science

- **Coursework:** Advanced Algorithms, Distributed Systems, Database Systems, Operating Systems, Computer Networks

Osmania University, Hyderabad **Jul 2017 - May 2021**

Bachelor of Technology, Computer Science and Engineering

- **Coursework:** Data Structures, Algorithms, Operating Systems, Computer Architecture, Database Management

EXPERIENCE

University Of Illinois at Chicago, Rehab Robotics Lab **Feb 2024 - Aug 2024**

Graduate Hourly (Software Engineer) *Chicago, USA*

- Implemented multi-objective Bayesian optimization (MOBO) with BoTorch/GPyTorch achieving 20% faster convergence, demonstrating strong algorithm design and optimization skills for robotics parameter tuning.
- Built end-to-end data pipeline processing real-time sensor streams from hardware (TwinCAT PLC, Arduino) via UDP sockets and serial protocols, applying systems programming and network communication concepts.
- Developed production-quality FastAPI dashboard with WebSocket for real-time visualization of Gaussian processes and 3D Pareto fronts, following software engineering best practices including 95% test coverage, type safety (mypy), and comprehensive code reviews.
- Optimized compute-intensive algorithms by profiling bottlenecks and applying vectorized NumPy operations with efficient data structures, improving performance by 40% through algorithmic improvements.

Deloitte **Sep 2021 - Aug 2023**

Analyst (Software Engineer) *Hyderabad, India*

- Designed and built enterprise integration platform using Java Spring Boot and Salesforce syncing 1.2M+ CRM records daily, implementing RESTful microservices with OAuth2 authentication, rate limiting, and comprehensive error handling for Fortune 500 client.
- Optimized system performance by 65% through database query optimization (SOQL indexing), Redis caching layer for frequently accessed data, and connection pooling (pgBouncer), demonstrating strong debugging and performance tuning skills.
- Built event-driven architecture with AWS Lambda, SQS, and SNS processing 500K+ events/day with 99.5% reliability, implementing circuit breakers (Hystrix), retry logic, dead-letter queues, and monitoring dashboards (Splunk/CloudWatch).
- Automated CI/CD pipeline reducing deployment time from 2 hours to 15 minutes using Jenkins and Salesforce DX, implemented automated testing (85% code coverage) and collaborated cross-functionally through code reviews and technical design discussions.

PROJECTS

Distributed Key-Value Store

- Built distributed database implementing Raft consensus protocol for leader election and log replication, demonstrating understanding of distributed systems, fault tolerance, and consistency models.
- Designed high-performance network layer using epoll for async I/O and custom binary protocol, achieving sub-millisecond latency through efficient data structures (hash tables, LRU cache) and memory management.
- Implemented production features including write-ahead logging (WAL), snapshot-based recovery, and data partitioning using consistent hashing across multi-node cluster.

Search Engine with Web Crawler

- Developed distributed web crawler using RabbitMQ for task distribution across multiple workers, implementing politeness policies (robots.txt, rate limiting) and URL deduplication with Bloom filters.
- Built inverted index with TF-IDF ranking and implemented PageRank algorithm for relevance scoring, achieving sub-200ms query latency on indexed documents using Elasticsearch.
- Optimized crawler scalability through multithreading, async I/O, and efficient data structures, processing thousands of pages per minute with minimal resource usage.

Real-Time Chat Application

- Built scalable chat system supporting 5,000+ concurrent connections using WebSocket protocol and Go's goroutines, implementing message persistence, user presence tracking, and typing indicators with sub-50ms message delivery.
- Designed message queue with Redis Streams for reliable delivery and implemented pagination with PostgreSQL for chat history, demonstrating understanding of event-driven architecture and database design patterns.
- Implemented authentication with JWT tokens, rate limiting per user (100 msg/min), and horizontal scalability through Redis pub/sub enabling multiple server instances to share connection state.

HTTP Load Balancer

- Implemented Layer 7 load balancer from scratch supporting multiple load-balancing algorithms (round-robin, least connections, weighted) with active health checks and automatic failover for backend servers.
- Built TCP connection pooling and HTTP request parsing using Python sockets, achieving 10,000+ requests/sec throughput with proper connection reuse and timeout handling.
- Designed circuit breaker pattern for fault isolation and implemented metrics collection (latency, error rates, connection counts) demonstrating production-ready monitoring and reliability patterns.

Achievements

- **LeetCode:** Solved 350+ problems focusing on data structures, algorithms, and system design patterns
- **Open Source:** Active contributor to Python and systems programming projects on GitHub
- **Technical Skills:** Strong problem-solving ability with deep understanding of computer science fundamentals